**1** Double glazing can be used for sound insulation. Double-glazed windows consist of two panes of glass with air in the space between them. Manufacturers reduce the air pressure in the space between the panes of glass to reduce the amplitude of sound transmitted through the window.

It is suggested that the amplitude $A$ of sound transmitted through a double-glazed window is related to the air pressure $p$ in the space between the panes by the equation

$$A = k\sqrt{p}$$

where $k$ is a constant.

Fig. 1.1 shows a laboratory model of a double-glazed window. It consists of two panes of glass. There is a tube connected to the space between the two panes so that air may be removed.

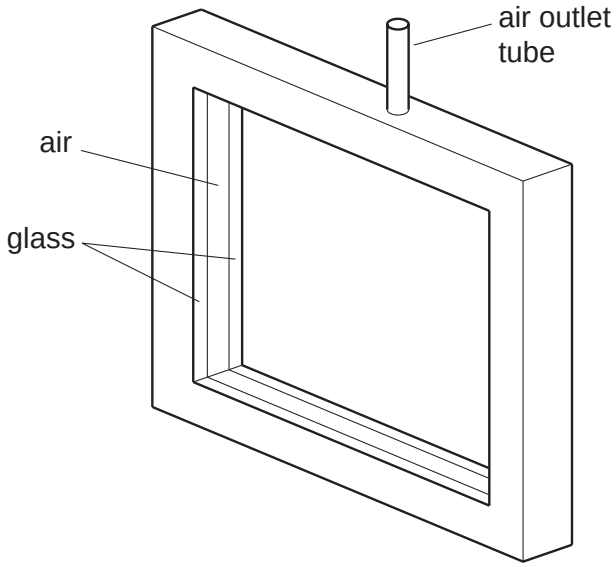


**Fig. 1.1**

Design a laboratory experiment to investigate whether $A$ is related to $p$ as indicated in the above equation when $p$ is reduced. You should draw a diagram showing the arrangement of your equipment. In your account you should pay particular attention to

**(a)** the procedure to be followed,

**(b)** how the air pressure between the panes of glass would be measured,

**(c)** how the amplitude of the sound would be measured,

**(d)** the control of variables,

**(e)** how the data would be analysed,

**(f)** any safety precautions that you would take.

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